

Cloud VM Infrastructure Monitoring Using Zabbix

Introduction to Zabbix

Zabbix is an open-source enterprise monitoring platform designed to monitor the performance, availability, and overall health of IT infrastructure. It enables organizations to monitor networks, servers, virtual machines, cloud services, databases, websites, applications, and other IT resources through a centralized web-based interface.

The platform gathers real-time metrics such as CPU utilization, memory usage, disk space, network traffic, service status, and system availability. This information is stored in a database and presented through graphs, dashboards, reports, and visual maps. These features help administrators quickly identify issues, analyze performance trends, and plan future resource requirements.

One of the key strengths of Zabbix is its flexible alerting mechanism. Users can define triggers for conditions such as high CPU utilization, low disk space, or server outages. Whenever a defined condition is met, Zabbix automatically sends notifications through email or other supported communication channels, allowing administrators to respond promptly and minimize downtime.

Zabbix supports multiple monitoring techniques, including agent-based monitoring, SNMP, IPMI, JMX, polling, trapping, and custom scripts. It is capable of monitoring various operating systems such as Linux, Windows, macOS, and Unix-based platforms, along with virtualization technologies like VMware and Hyper-V, as well as network devices including Cisco routers and switches.

Zabbix was developed by Alexei Vladishev in 2001, and its first stable release became available in 2004. Today, it is maintained and supported by Zabbix LLC (SIA Zabbix), headquartered in Riga, Latvia. Due to its scalability, flexibility, and open-source nature, Zabbix is widely adopted by both small businesses and large enterprises. It also offers API support and integration capabilities, making it suitable for customized monitoring environments.

Although Zabbix provides a wide range of powerful monitoring features, its initial configuration can be challenging for beginners and advanced implementations may require scripting knowledge. For improved visualization and reporting, many organizations integrate Zabbix with tools such as Grafana or Splunk. Overall, Zabbix plays a vital role in improving infrastructure visibility, maintaining system reliability, and enabling faster responses to operational issues.

Importance and Key Features of Zabbix

Zabbix is a centralized infrastructure monitoring platform that enables organizations to monitor servers, virtual machines, applications, cloud services, and network devices in real time. It helps administrators identify potential issues before they become critical, reducing downtime and improving overall system performance. Through centralized monitoring and intelligent alerting, Zabbix enhances the availability, reliability, and stability of critical IT infrastructure.

Key Features of Zabbix

Some of the key features of Zabbix are:

- **Real-Time Monitoring:** Continuously monitors servers, applications, and network devices.
- **Resource Utilization Tracking:** Tracks CPU usage, memory consumption, disk space, and network performance.
- **Availability Monitoring:** Monitors the operational status of systems and services.
- **Automated Alerting:** Sends notifications based on predefined triggers and threshold values.
- **Interactive Dashboards:** Provides a centralized view of infrastructure health and performance.
- **Reporting and Analytics:** Offers graphs, reports, and historical data for performance analysis and trend monitoring.
- **Multiple Monitoring Methods:** Supports agent-based, SNMP-based, and template-based monitoring.
- **Cloud and Virtual Environment Monitoring:** Monitors cloud resources, virtual machines, and virtual environments.
- **Centralized Management:** Collects and manages monitoring data from multiple systems through a single platform.

These features make Zabbix a scalable, flexible, and cost-effective monitoring solution for both small and large IT environments. As a result, it is widely used to improve infrastructure reliability, enhance operational efficiency, and support business continuity.

Zabbix Architecture

The Zabbix architecture is built to deliver efficient, scalable, and centralized monitoring of IT infrastructure. It consists of several interconnected components that work together to collect, process, store, and present monitoring data. This modular design allows Zabbix to monitor a wide variety of devices, applications, and services across different IT environments.

Main Components of Zabbix Architecture

1. Zabbix Server

The Zabbix Server is the central component of the monitoring system. It collects data from monitored devices, processes the received information, evaluates triggers, generates alerts, and stores monitoring data in the database.

2. Zabbix Agent

The Zabbix Agent is installed on monitored hosts, such as servers and virtual machines. It collects system metrics, including CPU usage, memory utilization, disk space, network activity, and running services, then sends this data to the Zabbix Server.

3. Database

The database stores configuration information, collected metrics, event logs, alerts, and historical monitoring data. Zabbix commonly supports databases such as MySQL, PostgreSQL, and MariaDB.

4. Web Frontend

The Web Frontend provides an easy-to-use graphical interface for administrators. It allows users to configure hosts, create dashboards, view reports, analyze performance graphs, and manage alerts through a web browser.

5. Zabbix Proxy (Optional)

The Zabbix Proxy serves as an intermediary between monitored devices and the Zabbix Server. It gathers monitoring data from remote locations and forwards it to the server, helping reduce network traffic and improve overall scalability.

Working of the Architecture

1. The Zabbix Agent gathers performance and availability data from the monitored systems.
2. The collected information is transmitted to the Zabbix Server either directly or through a Zabbix Proxy.
3. The Zabbix Server processes the received data and stores it in the database.
4. The server evaluates configured triggers and generates alerts whenever predefined conditions are satisfied.
5. Administrators access monitoring data through the Web Frontend, where information is presented using dashboards, graphs, reports, and alert notifications.

The Zabbix architecture provides centralized monitoring, efficient data processing, and real-time visibility into the health, availability, and performance of the entire IT infrastructure.

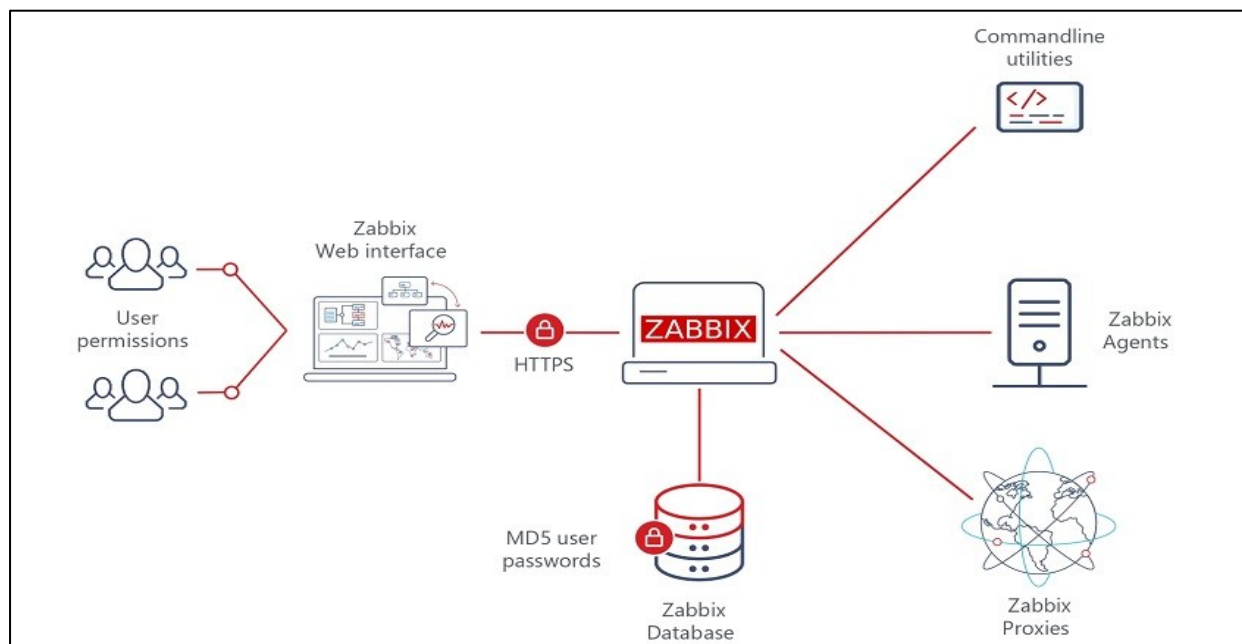


Figure: Architecture of Zabbix

Monitoring Methods in Zabbix

Zabbix supports a variety of monitoring methods, enabling organizations to monitor different types of IT infrastructure efficiently. Based on the type of device or service, Zabbix can gather monitoring data using agents, network protocols, or preconfigured templates.

Monitoring Method	Description	Examples
Agent-Based Monitoring	Uses the Zabbix Agent installed on the monitored host to collect detailed system metrics.	CPU Utilization, Memory Usage, Disk Space Usage, System Uptime
SNMP Monitoring	Uses the SNMP protocol to monitor network devices without requiring an agent installation.	Routers, Switches, Firewalls, Printers
ICMP Monitoring	Uses ping requests to verify the availability and reachability of hosts.	Host Availability, Network Connectivity
Template-Based Monitoring	Uses predefined templates containing monitoring items, triggers, and graphs for fast and consistent configuration.	Linux by Zabbix Agent Template

Table: Monitoring methods

Triggers, Alerts, and Dashboards

Zabbix uses triggers to identify abnormal system conditions and automatically generate alerts when predefined threshold values are exceeded. Common trigger conditions include high CPU utilization, excessive system load, low disk space, and host unavailability. When an issue is detected, Zabbix can send notifications to administrators through email or other configured notification channels.

Zabbix also offers centralized dashboards that provide real-time visibility into the health and performance of IT infrastructure. The dashboard displays important information such as:

- Host Availability
- CPU and Memory Utilization
- Active Problems
- Infrastructure Status
- Performance Graphs
- Alert Summaries

In addition, historical monitoring data and reports enable administrators to analyze performance trends, troubleshoot system issues, and support effective capacity planning.



Figure: Zabbix Dashboard

Mini Project: Cloud VM Infrastructure Monitoring Using Zabbix

Project Objective

The objective of this project was to design and implement a centralized infrastructure monitoring solution using Zabbix in a cloud environment. The system was built to monitor multiple virtual machines, identify abnormal system behaviour, generate alerts based on predefined conditions, and send email notifications to administrators. This project demonstrates how Zabbix enhances infrastructure visibility, improves system reliability, and enables faster incident response.

Infrastructure Setup

The monitoring environment was deployed on VMware Cloud Director (VCD). The infrastructure included three virtual machines: one Zabbix Server and two monitored client systems.

Virtual Machine	IP Address	Purpose
Zabbix Server	192.168.0.2	Central monitoring server that collects monitoring data, processes alerts, and provides dashboards.
Client-1	192.168.0.3	Linux virtual machine monitored through Zabbix.
Client-2	192.168.0.4	Linux virtual machine monitored through Zabbix.

Table: Infrastructure Setup

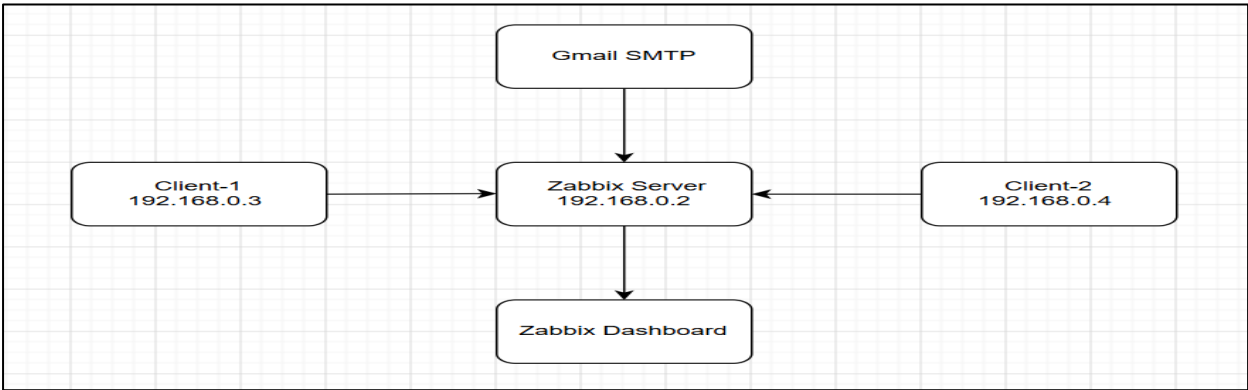


Figure: Project Architecture

Zabbix Server Configuration

The Zabbix Server was configured with the core components required for monitoring, data storage, and visualization. The following services were installed and configured:

- **MySQL Database** – Used to store monitoring data and configuration details.
- **Apache Web Server** – Hosted the Zabbix web interface.
- **Zabbix Frontend** – Provided a graphical interface for monitoring and administration.
- **Zabbix Agent** – Collected system metrics from the monitored hosts.

The Zabbix web dashboard was accessible through the server's public IP address, enabling administrators to monitor system status, view alerts, and analyze monitoring data from a centralized interface.

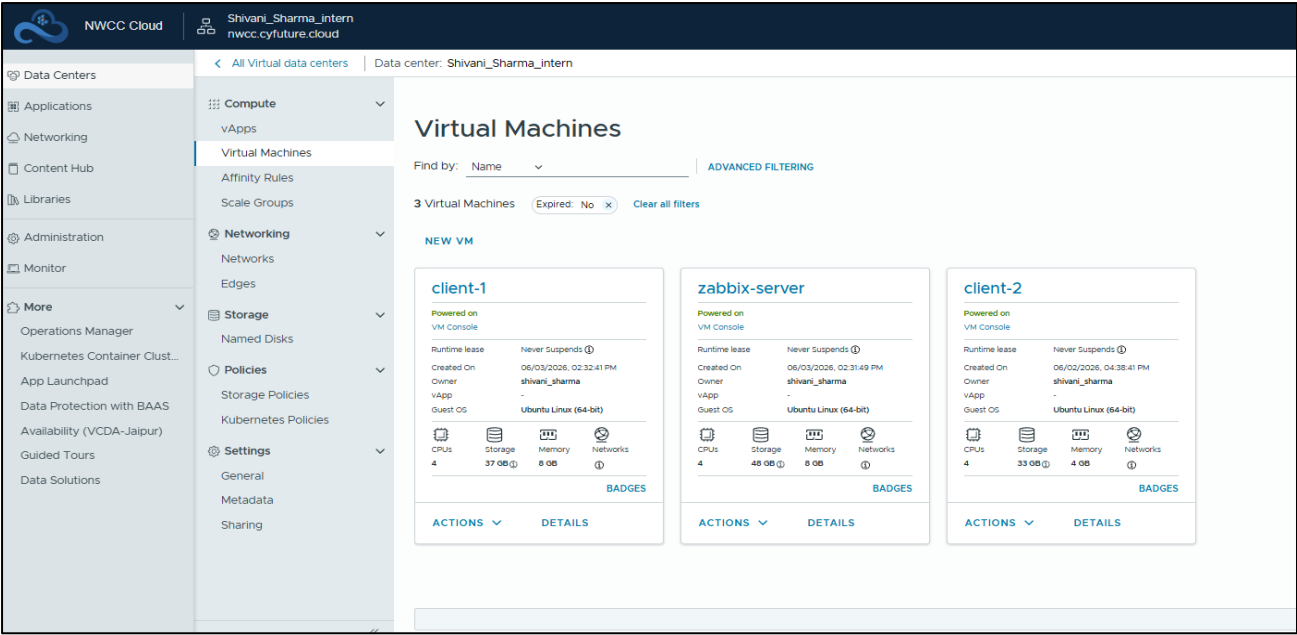


Figure: Infrastructure Setup

Zabbix Server Installation and Configuration

After deploying the virtual machines in VMware Cloud Director (VCD), the Zabbix Server was installed and configured on an Ubuntu 22.04 virtual machine with the IP address 192.168.0.2.

Server Details

Parameter	Value
Hostname	zabbix-server
Operating System	Ubuntu 22.04 LTS
IP Address	192.168.0.2
Purpose	Centralized Monitoring Server

Table: Server Details

Zabbix Server Installation and Configuration

Step	Task	Commands / Configuration	Purpose
Step 1	Update System Packages	apt update apt upgrade -y	Updates the system packages to their latest versions before starting the installation process.
Step 2	Install MySQL Server	apt install mysql-server -y systemctl status mysql	Installs the MySQL database server required by Zabbix to store monitoring data.

Step 3	Secure MySQL Installation	mysql_secure_installation	Secures the MySQL installation by setting the root password, removing anonymous users, disabling remote root login, deleting the test database, and reloading privileges.
Step 4	Create Zabbix Database	mysql -u root -pCREATE DATABASE zabbix CHARACTER SET utf8mb4 COLLATE utf8mb4_bin;CREATE USER 'zabbix'@'localhost' IDENTIFIED BY 'zabbix123';GRANT ALL PRIVILEGES ON zabbix.* TO 'zabbix'@'localhost';FLUSH PRIVILEGES;	Creates the Zabbix database and user account required for storing monitoring and configuration data.
Step 5	Install Zabbix Repository	wget https://repo.zabbix.com/zabbix/7.0/ubuntu/pool/main/z/zabbix-release/zabbix-release_latest_7.0+ubuntu22.04_all.debdpkg -i zabbix-release_latest_7.0+ubuntu22.04_all.debapt update	Adds the official Zabbix repository to the Ubuntu system and updates the package index.
Step 6	Install Zabbix Components	apt install zabbix-server-mysql zabbix-frontend-php zabbix-apache-conf zabbix-sql-scripts zabbix-agent -y	Installs the Zabbix Server, frontend, Apache configuration, SQL scripts, and agent packages.
Step 7	Import Database Schema	zcat /usr/share/zabbix-sql-scripts/mysql/server.sql.gz mysql -u zabbix -p zabbix	Imports the default database schema, tables, and triggers required by the Zabbix Server.
Step 8	Configure Zabbix Server	Edit /etc/zabbix/zabbix_server.confDBPassword=zabbix123	Configures the database password, allowing the Zabbix Server to connect to the MySQL database.
Step 9	Restart and Enable Services	systemctl restart zabbix-serversystemctl restart apache2systemctl restart zabbix-agentsystemctl enable zabbix-serversystemctl enable apache2systemctl enable zabbix-agent	Restarts all required services and enables them to start automatically after every system boot.
Step 10	Verify Services	systemctl status zabbix-serversystemctl status apache2ss -tulpn grep 10051	Verifies that the Zabbix Server, Apache services, and port 10051 are operating correctly.

Table: Zabbix Server Installation and Configuration

Step 11: Access Zabbix Web Interface

The Zabbix web interface was accessed through a web browser.

<http://PUBLIC-IP/zabbix>

The installation wizard was completed by:

- Verifying the PHP requirements and configuring the database connection.
- Providing the server information and completing the frontend setup.

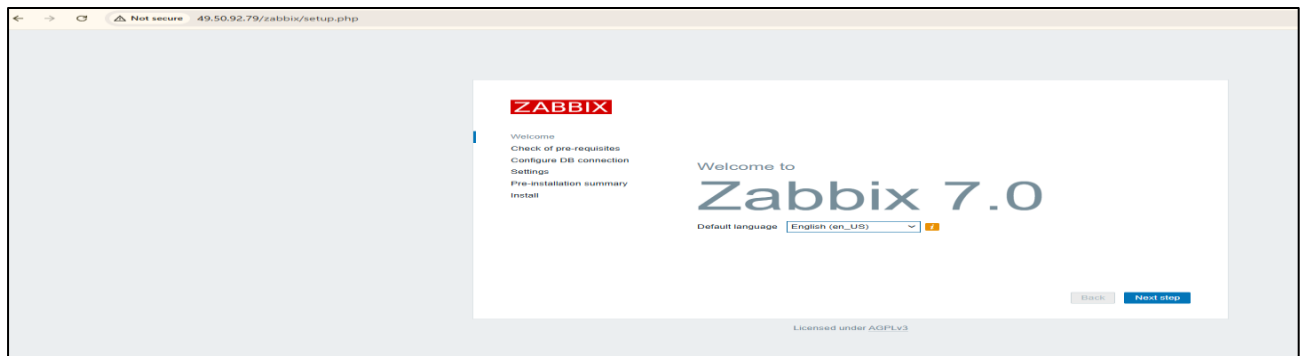


Figure: Access Zabbix Web Interface

Step 12: Login to Zabbix Dashboard

After completing the installation, the Zabbix dashboard was accessed using the default login credentials.

Username: Admin

Password: zabbix

For security purposes, the default password was changed immediately after the first successful login.

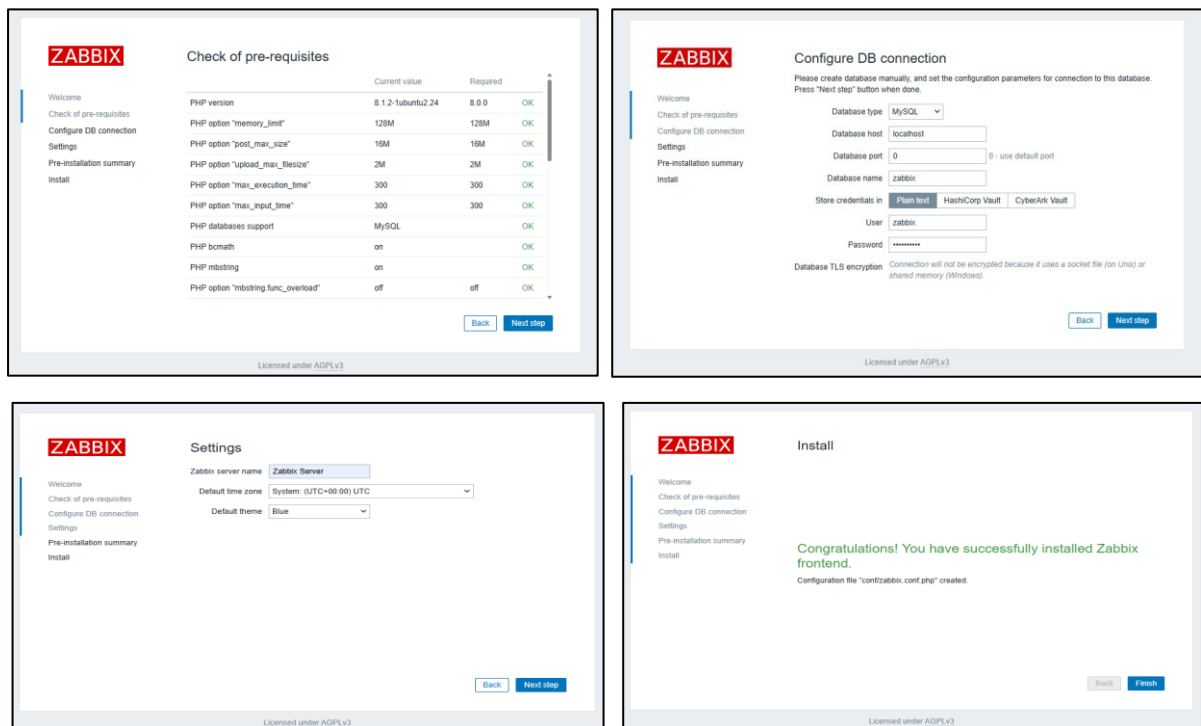


Figure: Login to Zabbix Dashboard

Result

The Zabbix Server was successfully installed and configured on Ubuntu 22.04. It was successfully connected to the MySQL database, provided web-based monitoring through the Zabbix frontend, and was ready to receive monitoring data from the Client-1 and Client-2 virtual machines.

Agent Installation and Configuration

After the successful deployment and configuration of the Zabbix Server, Zabbix Agents were installed on both monitored virtual machines (Client-1 and Client-2). The Zabbix Agent collects system metrics such as CPU utilization, memory usage, disk space usage, system load, and host availability, and sends this information to the Zabbix Server for centralized monitoring.

Monitored Virtual Machines

Host	IP Address	Purpose
Client-1	192.168.0.3	Monitored Linux VM
Client-2	192.168.0.4	Monitored Linux VM

Zabbix Agent Installation and Configuration (Client-1 and Client-2)

Step	Task	Commands / Configuration	Purpose
Step 1	Update Package Repository	Client-1: apt update Client-2: apt update	Updates the package repository before installing the required packages.
Step 2	Install Zabbix Agent	Client-1: apt install zabbix-agent -y Client-2: apt install zabbix-agent -y	Installs the Zabbix Agent on both client virtual machines.
Step 3	Verify Agent Installation	zabbix_agentd --version	Verifies that the Zabbix Agent has been installed successfully.
Step 4	Configure Agent Settings	Edit: /etc/zabbix/zabbix_agentd.conf	Configures communication between the Zabbix Server and the client agents.

	Client-1 Configuration	Server=192.168.0.2ServerActive=192.168.0.2Hostname=client-1	Configures Client-1 to communicate with the Zabbix Server.
	Client-2 Configuration	Server=192.168.0.2ServerActive=192.168.0.2Hostname=client-2	Configures Client-2 to communicate with the Zabbix Server.
Step 5	Restart Agent Service	Client-1: systemctl restart zabbix-agent Client-2: systemctl restart zabbix-agent	Restarts the Zabbix Agent service to apply the updated configuration
Step 6	Enable Service at Boot	Client-1: systemctl enable zabbix-agent Client-2: systemctl enable zabbix-agent	Ensures the Zabbix Agent starts automatically whenever the system boots.
Step 7	Verify Agent Status	systemctl status zabbix-agent	Confirms that the Zabbix Agent service is active and running correctly.
Step 8	Verify Agent Port	ss -tulpn grep 10050	Verifies that port 10050 is listening for connections from the Zabbix Server.

Table: Zabbix Agent Installation and Configuration

Step 9: Add Hosts in Zabbix Dashboard

After configuring the Zabbix Agents, both client virtual machines were added to the Zabbix Dashboard for monitoring.

Navigation:

Data Collection

→ Hosts

→ Create Host

Parameter	Client-1	Client-2
Host Name	client-1	client-2
Host Group	Linux Servers	Linux Servers
IP Address	192.168.0.3	192.168.0.4
Template Applied	Linux by Zabbix Agent	Linux by Zabbix Agent

Table: Add Hosts

Step 10: Verify Successful Communication

After adding the hosts, the **Availability** column on the **Hosts** page was checked to verify connectivity.

Navigation:

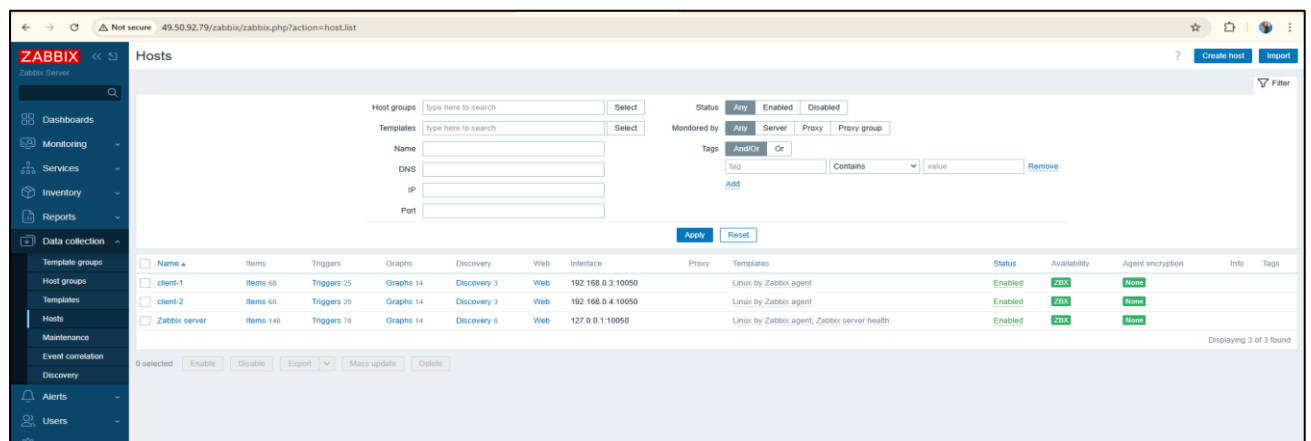
Data Collection

→ Hosts

A green **ZBX** indicator was displayed for both **Client-1** and **Client-2**.

This confirmed that:

- Agent communication was established successfully.
- Monitoring data was being collected correctly.
- Both hosts were actively connected to the Zabbix Server.



Name	Items	Triggers	Graphs	Discovery	Web	Interface	Proxy	Templates	Status	Availability	Agent encryption
client-1	Items 68	Triggers 25	Graphs 14	Discovery 3	Web	192.168.0.3:10050		Linux by Zabbix agent	Enabled	ZBX	None
client-2	Items 68	Triggers 25	Graphs 14	Discovery 3	Web	192.168.0.4:10050		Linux by Zabbix agent	Enabled	ZBX	None
Zabbix server	Items 146	Triggers 78	Graphs 14	Discovery 6	Web	127.0.0.1:10050		Linux by Zabbix agent, Zabbix server health	Enabled	ZBX	None

Figure: Verification of Successful Communication

Result

The Zabbix Agents were successfully installed and configured on both Client-1 and Client-2 virtual machines. Both monitored hosts successfully established communication with the Zabbix Server and started transmitting CPU usage, memory utilization, disk usage, load average, and host availability metrics, enabling centralized monitoring of the cloud infrastructure.

Monitoring Configuration and Dashboard Development

A custom dashboard named "Cloud VM Infrastructure Monitoring" was created to provide a centralized view of the monitored infrastructure. The dashboard displayed essential information, including host availability, CPU utilization, active alerts, infrastructure status, and overall system health.

Dashboard Components: VM Availability, Current Alerts, CPU Utilization, Infrastructure Status, Clock Widget

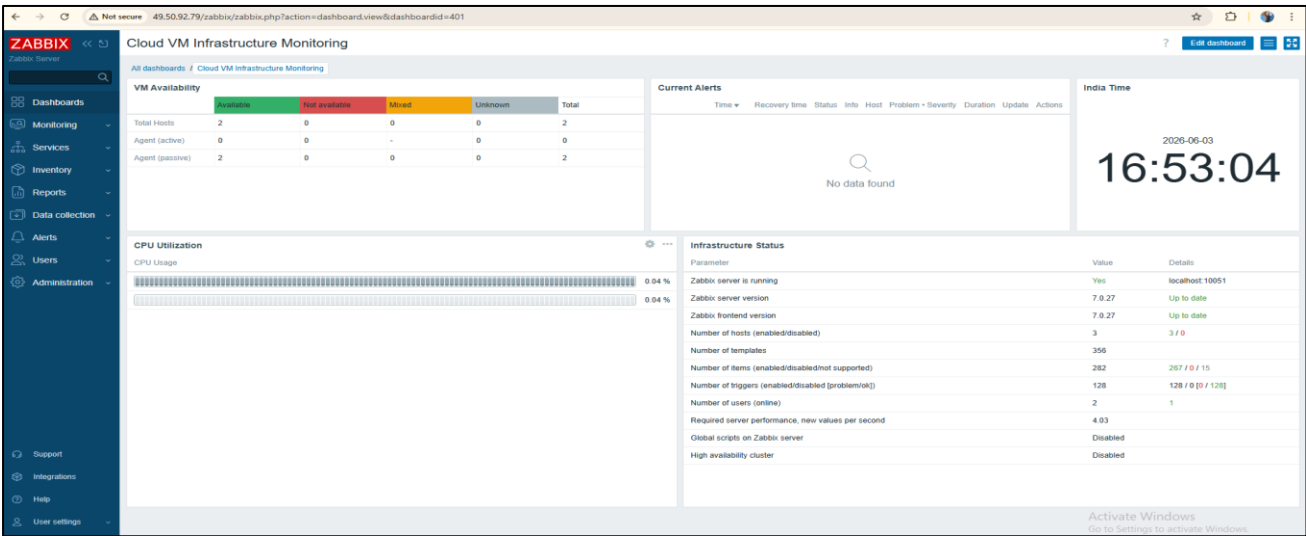


Figure: Custom Zabbix Monitoring Dashboard

The monitoring configuration was designed to track important system parameters, including:

- **CPU Monitoring:** CPU utilization and load average
- **Memory Monitoring:** Available memory and memory usage
- **Disk Monitoring:** Disk usage and available space
- **Host Monitoring:** Host availability and agent status
- **Security Monitoring:** SSH failed login attempts and file integrity monitoring.

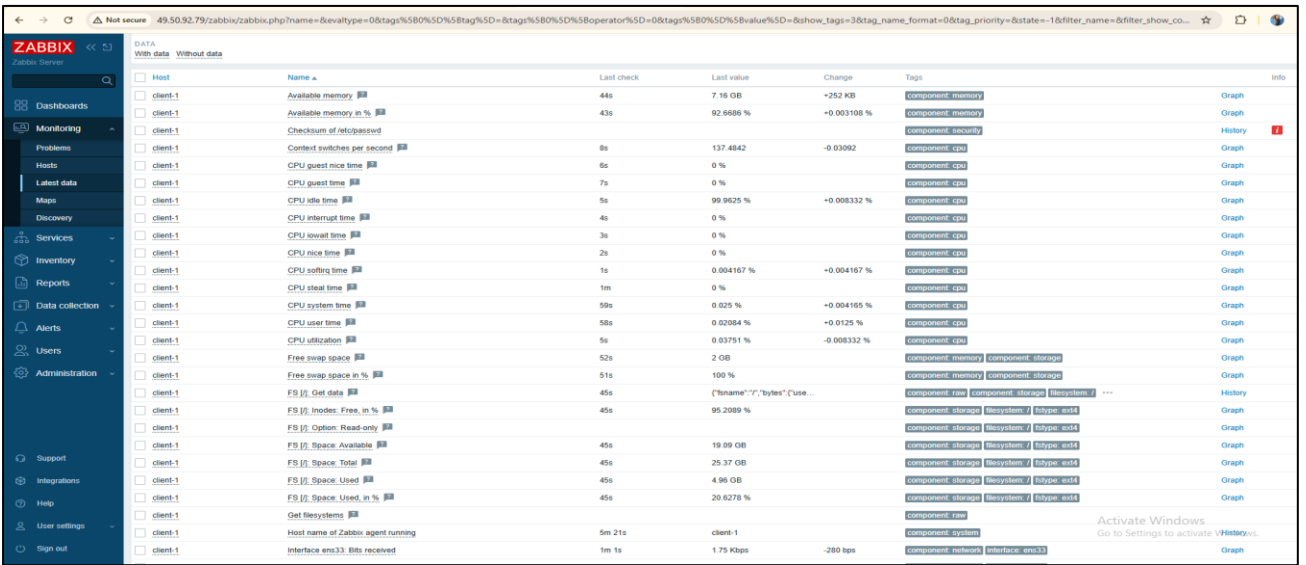


Figure: Latest Monitoring Data in Zabbix

To detect abnormal system behavior, triggers were configured for key events, including:

- **High CPU Usage**
- **High Load Average**
- **Low Disk Space**
- **Host Unreachable**

These triggers allowed Zabbix to generate alerts and notify administrators whenever the configured threshold values were exceeded.

CPU Alert Demonstration

A CPU stress test was performed using the following command:

stress --cpu 4 --timeout 300

Results:

- CPU utilization increased significantly.
- The configured trigger was activated.
- An alert was displayed on the dashboard.
- An email notification was sent to the administrator.

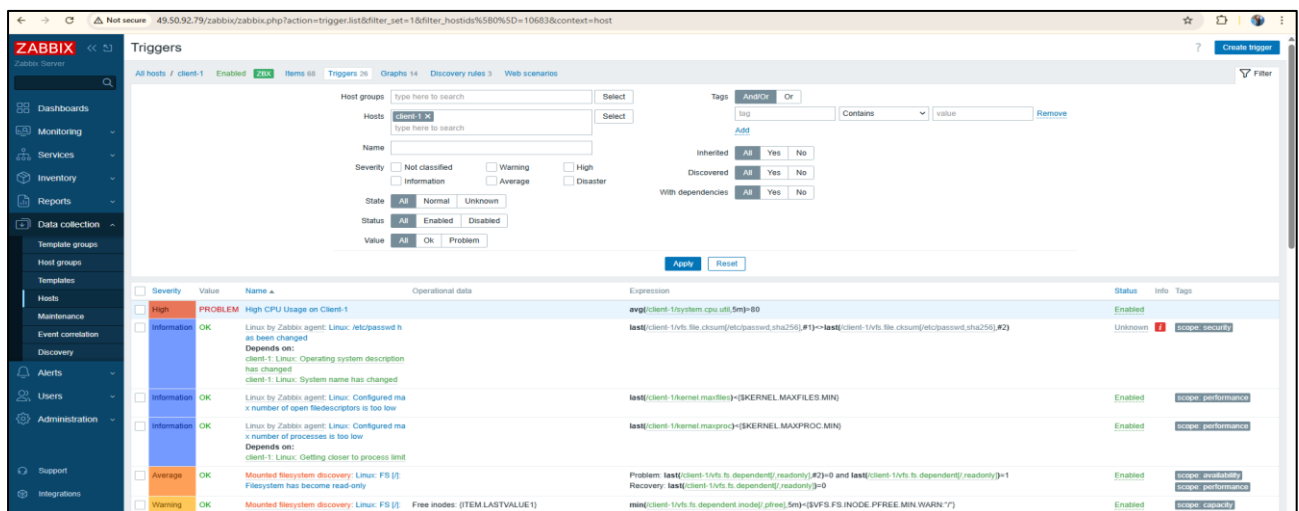
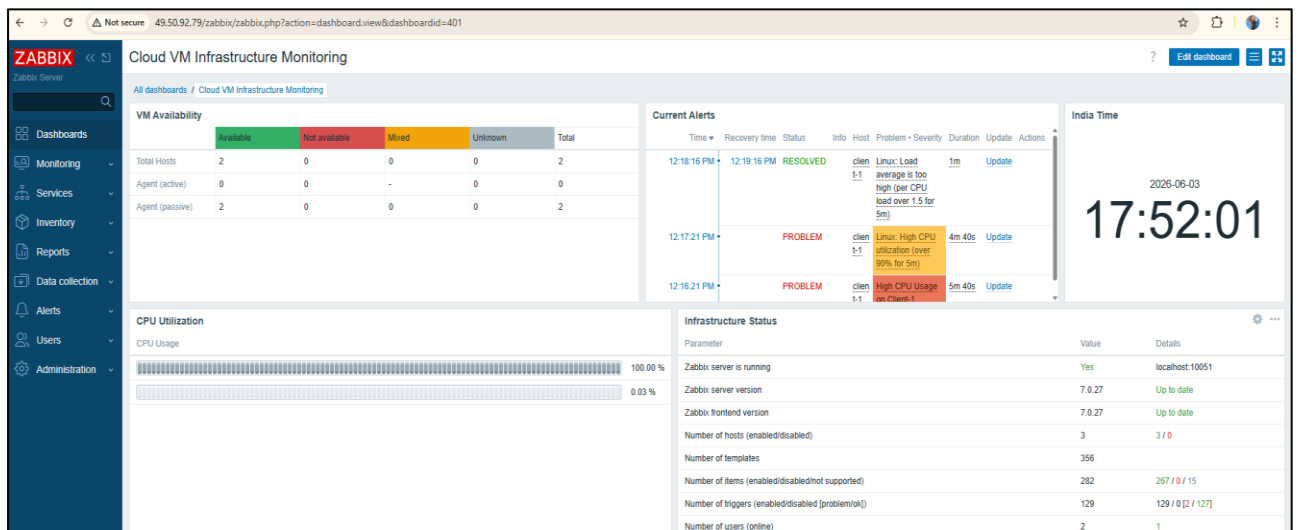


Figure: CPU Alert Demonstration

SSH Failed Login Monitoring

To incorporate a cybersecurity aspect into the project, SSH failed login monitoring was implemented.

A log monitoring item was configured:

log[/var/log/auth.log,Failed password]

The system monitored failed SSH authentication attempts and recorded the events in Zabbix.

This implementation demonstrated the security monitoring capabilities of the Zabbix platform.



The screenshot shows the Zabbix web interface with the 'client-1: SSH Failed Login Detection' history view. The table lists several failed login attempts for the user 'fakeuser' from IP 127.0.0.1 on ports 55000 and 54998.

Timestamp	Local time	Value
2026-06-03 04:02:16 PM	Jun 3 16:02:12	client-1 sshd[5515]: Failed password for invalid user fakeuser from 127.0.0.1 port 55000 ssh2
2026-06-03 04:02:06 PM	Jun 3 16:02:05	client-1 sshd[5515]: Failed password for invalid user fakeuser from 127.0.0.1 port 55000 ssh2
2026-06-03 04:02:06 PM	Jun 3 16:01:58	client-1 sshd[5515]: Failed password for invalid user fakeuser from 127.0.0.1 port 55000 ssh2
2026-06-03 04:01:26 PM	Jun 3 15:44:20	client-1 sshd[4817]: Failed password for invalid user fakeuser from 127.0.0.1 port 54998 ssh2
2026-06-03 04:01:26 PM	Jun 3 15:44:13	client-1 sshd[4817]: Failed password for invalid user fakeuser from 127.0.0.1 port 54998 ssh2
2026-06-03 04:01:26 PM	Jun 3 15:44:11	client-1 sshd[4817]: Failed password for invalid user fakeuser from 127.0.0.1 port 54998 ssh2

Figure: SSH Failed Login Monitoring

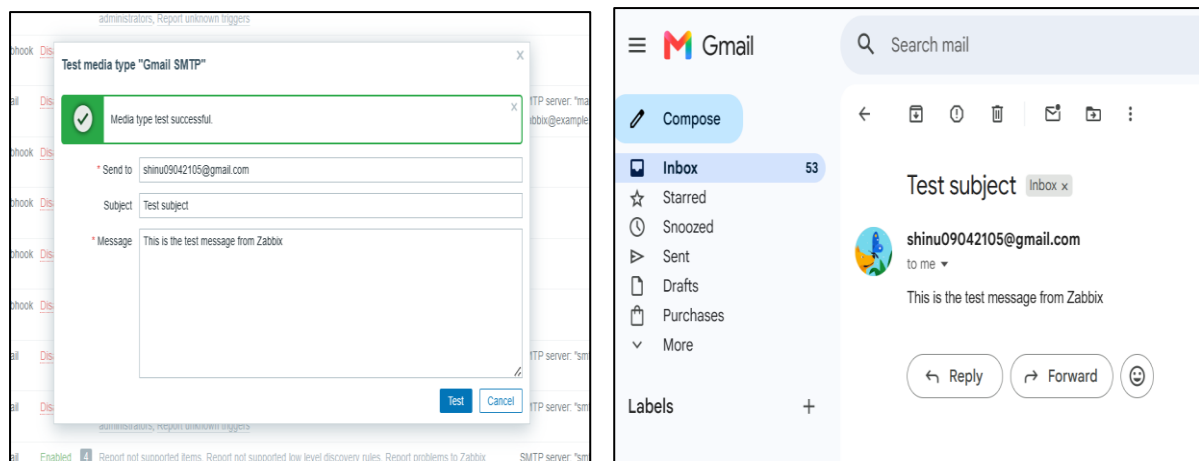
Email Notification Integration

Gmail SMTP was integrated with the Zabbix monitoring server to enable automated email notifications.

The configuration included:

- **SMTP Server:** smtp.gmail.com
- **Port:** 587
- **Security:** STARTTLS
- **Authentication:** Gmail App Password

Whenever a trigger changed to a problem state, Zabbix automatically sent email notifications to the configured recipient.



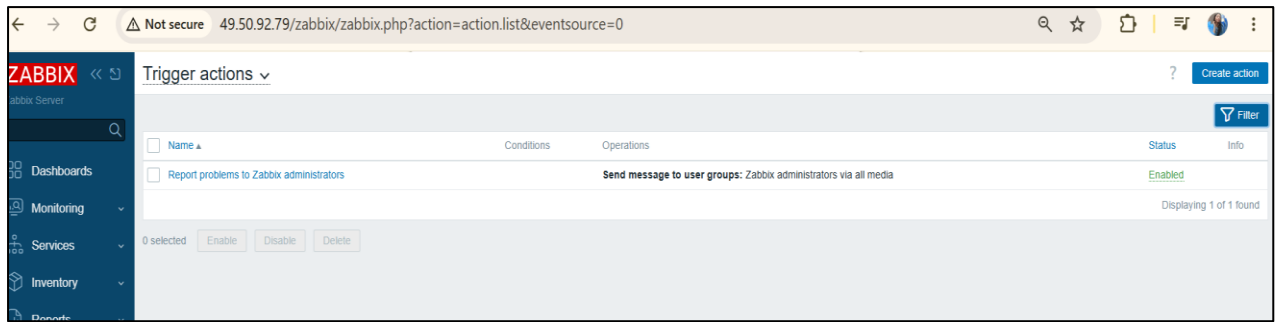


Figure: Email Notification Integration

Alert Testing

The configured alerting mechanism was tested to verify its functionality.

Problem Notification

A high CPU utilization condition was intentionally generated to trigger an alert.

Result: Alert generated. Email received.

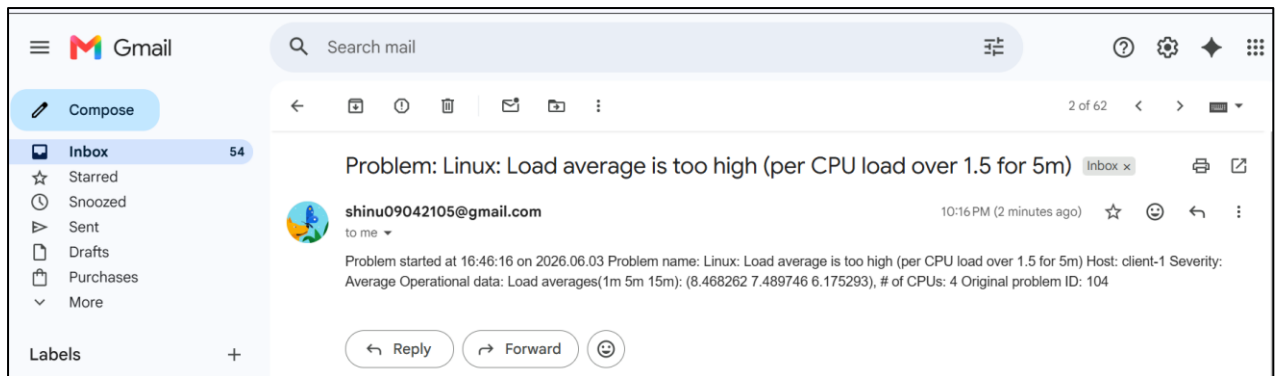


Figure: Problem Notification

Recovery Notification

After the CPU utilization returned to the normal threshold:

- Problem was resolved.
- A recovery email notification was received.

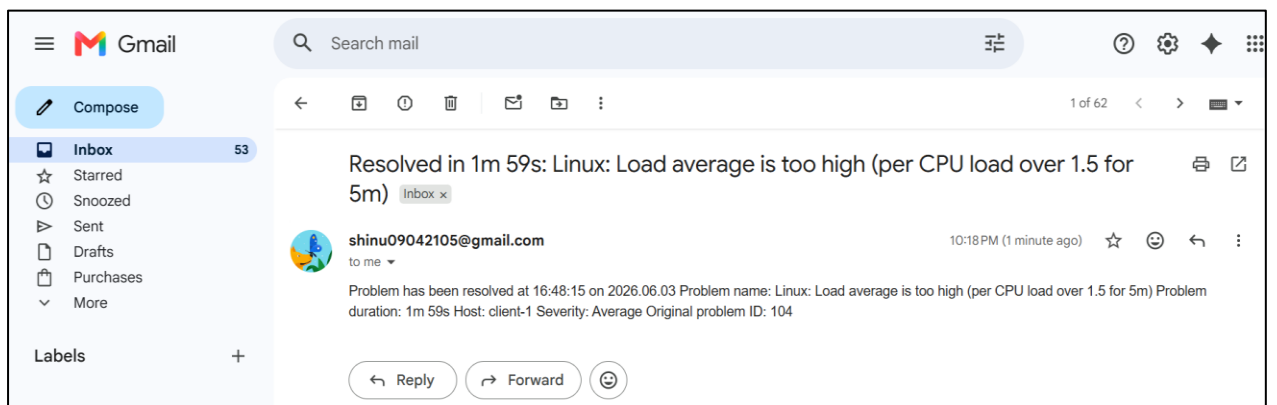


Figure: Recovery Notification

Problems Faced and Resolutions

Problem	Error / Cause	Resolution
Host Not Found Error	Error: host [client-1] not found Cause: Hostname mismatch between the Zabbix Agent and the Zabbix Server.	Updated the Hostname parameter in zabbix_agentd.conf and matched it with the hostname configured in Zabbix.
SSH Monitoring Not Supported	Error: Accessible only as active check. Cause: The log monitoring item required active checks.	Modified the monitoring item to use active checks.
Permission Denied for auth.log	Error: Cannot open file /var/log/auth.log. Cause: The Zabbix user did not have permission to read the authentication log.	Added the zabbix user to the adm group and restarted the Zabbix Agent service.
SMTP Login Denied	Error: Login denied. Cause: Gmail blocked authentication using a regular account password.	Enabled Two-Factor Authentication (2FA) and generated a Gmail App Password for SMTP authentication.
Email Notifications Not Received	Cause: The media configuration was not associated with the administrator account.	Configured Gmail SMTP for the Admin user and enabled the required trigger actions for email notifications.

Table: Problems Faced and Resolutions

Project Repository

<https://github.com/2121-shivi/Cloud-VM-Infrastructure-Monitoring-Using-Zabbix->

Project Outcomes and Conclusion

The project was successfully completed by implementing a Cloud VM Infrastructure Monitoring and Alerting System using Zabbix. The solution provided centralized monitoring of virtual machines, real-time performance monitoring, interactive dashboard visualization, and automated email notifications. The implementation demonstrated that Zabbix is a reliable and efficient monitoring platform for proactive issue detection, improved infrastructure visibility, and effective cloud infrastructure management.